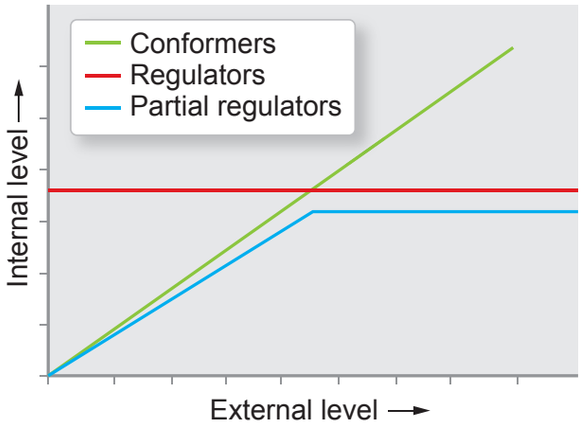


Fig. 10.9 Organisms' response to environmental stressors

The types of responses observed are

- **Regulate:** Some organisms are able to maintain homeostasis by physiological means which ensures constant body temperature, ionic / osmotic balance. Birds, mammals and a few lower vertebrate and invertebrate species are capable of such regulation.
- **Conform**: Most animals cannot maintain a constant internal environment. Their body temperature changes with the ambient temperature. In aquatic animals like fishes, the osmotic concentration of the body fluids changes with that of the ambient water osmotic concentration. Such animals are called **Conformers**. In case of extreme condition, the inhabitants relocate themselves as in migration.
- **Migrate:** Organisms tend to move away temporarily from a stressful habitat to a new, hospitable area and return when the stressful period is over. Birds migrate from Siberia to Vedanthangal in Tamilnadu to escape from the severe winter periods.
- **Suspend:** In certain conditions, if the organisms is unable to migrate, it may avoid the stress by becoming inactive. This is seen commonly in bears going into **hibernation** during winter. Some snails and fish go into **aestivation** to avoid summer related problems like heat and desiccation. Some lower animals suspend a certain phase of their life cycle, which is referred to as **diapause**.

10.6 Adaptations

In biology, **adaptation** is a dynamic evolutionary process that fits organisms to their environment and enhancing their evolutionary fitness. Adaptations can be a phenotypic or **adaptive trait** with a functional role in each individual organism that is maintained and has been evolved by natural selection. The adaptive traits may be structural adaptation, behavioural adaptation and physiological adaptation.

a) Structural adaptations

The external and internal structures of animals can help them to **adapt better to their environment**. Some of the most common examples are mammals growing thicker fur to survive freezing climates. Some of the most attractive adaptations in nature occur for reasons of crypsis (e.g. camouflage) and mimicry. Cryptic animals are those which **camouflage perfectly** with their environment and are almost impossible to detect. Certain reptiles and insects such as chameleons and stick insects show this type of adaptation, which helps in prey capture or to evade from predators. Likewise, horse legs are suitable for fast running and adapted for grasslands and similar terrestrial environments.

b) Behavioural adaptations

Action and behaviour of animals are instinctive or learned. Animals develop certain behavioural traits or adaptations for survival. Fleeing from a predator, hiding during sleep, seeking refuge from climate change or moving to find different food sources are all behavioral adaptations. The two most characteristic forms of behavioral adaptations are migration and courtship. Migration allows the animals to find

XII Std Biology-Zoology Chapter-10 EM.indd 172 23/12/2022 13:07:07